Do speakers monitor their speech? Evidence from post-naming gazes to objects

Prior research showed that the time speakers spend gazing at objects before naming them reflects the difficulty of retrieving their names (Griffin, 2001; Meyer et al., 1998). Gaze usually shifts away from the object after phonological encoding and before phonetic encoding (Meyer & Van der Meulen, 2000). However, little is known about gaze after naming, despite theoretical claims that speakers monitor the fit of the produced label to the referent (Levelt, 1983). To investigate whether speakers monitor their object names for fit after word onset and whether there are age differences in monitoring, we analyzed data from a study that included 30 English-speaking adults (18-23 and 60-80 years old). The analysis plan was preregistered. Participants named 3 objects using the frame, "The A and the B are above the C," and we analyzed data for naming the objects in position B. If speakers monitor names for fit, we predicted that their post-name gaze durations would show effects of name agreement and word frequency, which were manipulated in the experiment. Given slower word recognition speed and greater cautiousness, we also predicted high probabilities and durations of gazes for older adults.

On 58% of trials, speakers gazed at objects after name onset. Participants were more likely to look back when word frequency, $\beta = -0.398$, $p = .025$, and name agreement, $\beta = -0.642$, $p < .001$, were lower (Fig. 1). This suggests that speakers were more likely to monitor labels that were more difficult to retrieve. When participants did look at objects after label onset, gaze durations were longer for objects with lower name agreement, $\beta = -152.24$, $p = .002$, especially for older adults, $\beta = 136.62$, $p = .049$. Gaze durations were marginally longer when word frequency was lower, $\beta = -79.63$, $p = .082$. See Figure 2 for an illustration of the time course. Work on spoken word recognition shows slower recognition of lower frequency words (e.g., Dahan et al., 2001) and it probably takes longer to confirm a label for an object with more variable word-to-concept mappings. Gaze durations after name onset likely reflect the speed of name recognition in the service of monitoring for accuracy.

In addition, analyses were conducted on the eye movements of 41 English-speaking parents spontaneously labeling objects while playing with their 12-24-month-old children to test whether monitoring extends to natural interactions and situations where attending to an interlocutor may be far more important to speakers than monitoring their own speech. In 63% of naming instances, parents looked at objects within 3 seconds after name onset. They were more likely to look when word frequency, $\beta = -0.167$, $p < .023$, and name agreement were lower, $\beta = -0.174$, $p < .012$, and less likely as object names were repeated, $\beta = -0.222$, $p < .019$. The interaction between object repetition and name agreement, $\beta = -0.166$, $p < .035$, was due to name agreement having a variable impact with repetition (Fig. 3). On the trials with gazes after onset, parents spent less time gazing at objects as they repeatedly named them, $\beta = -0.098$, $p = .055$. The frequency of an object's name had less of an impact as it was repeatedly named, $\beta = 0.029$, $p = .030$. Name agreement and the number of times they previously named objects from the same semantic category had no significant effect, nor were there further interactions.

Indices of lexical difficulty, which affect both word production and recognition, influenced the probability and duration of gazing at objects after articulation of their names began in both a controlled laboratory task and naturalistic parent–child interactions. Although older adults were equally likely to revisit objects, the duration of their gazes was longer particularly for objects with low name agreement, partially consistent with the idea that older adults are more cautious about their speech. Importantly, the natural naming data show that monitoring occurs even when speakers are engaged in socially demanding tasks (interacting with the child). Together, these studies provide strong evidence that speakers continue monitoring for object names for fit after articulation across the adult life span, in experimental and natural settings.

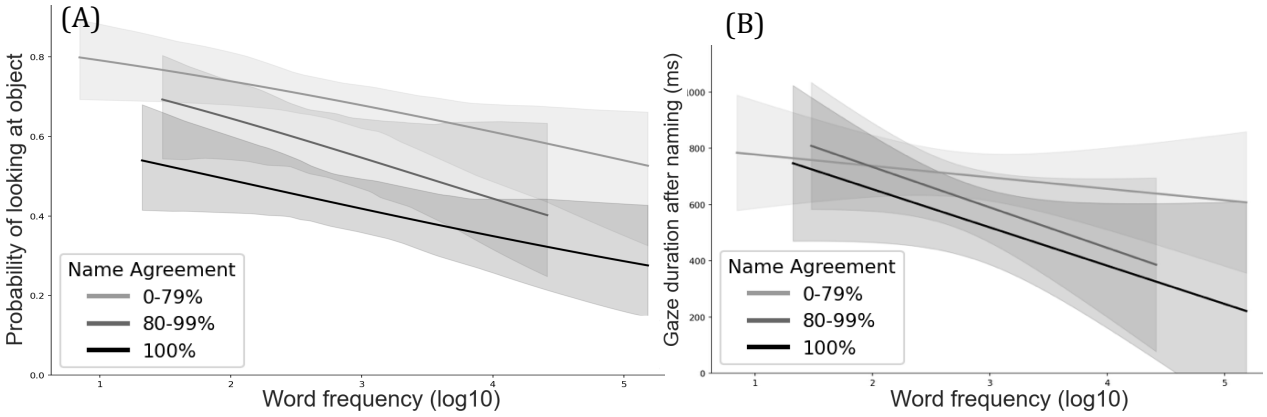

Figure 1. For Study 1, (A) Mean probability (95% CI) of looking at objects after naming them as a function of name agreement and word frequency. (B) Mean gaze duration (95% CI) when objects were gazed at as a function of name agreement and word frequency, collapsing over the significant effect of age group for simplicity.

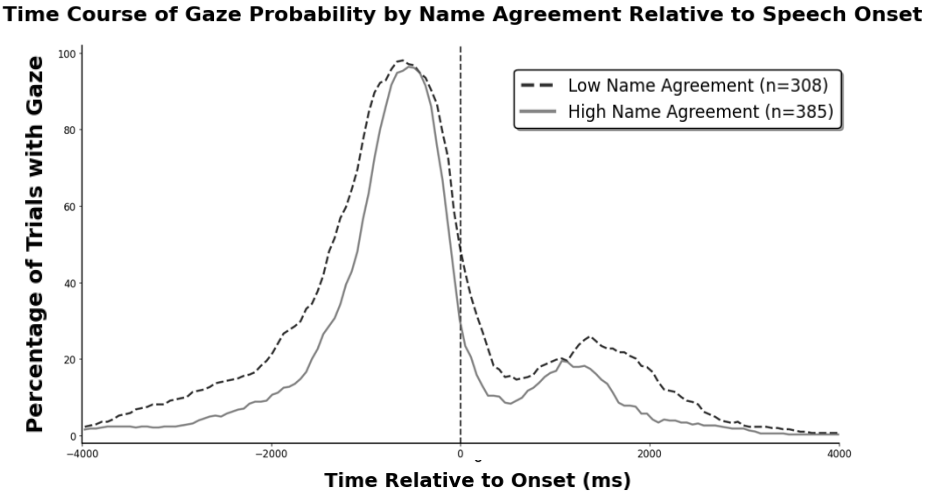

Figure 2: Study 1. Time course of gazes to target objects relative to name onset. For simplicity, data is collapsed over frequency and age. The second peak after name onset suggests that participants often looked away from the object after producing the name and then looked back.

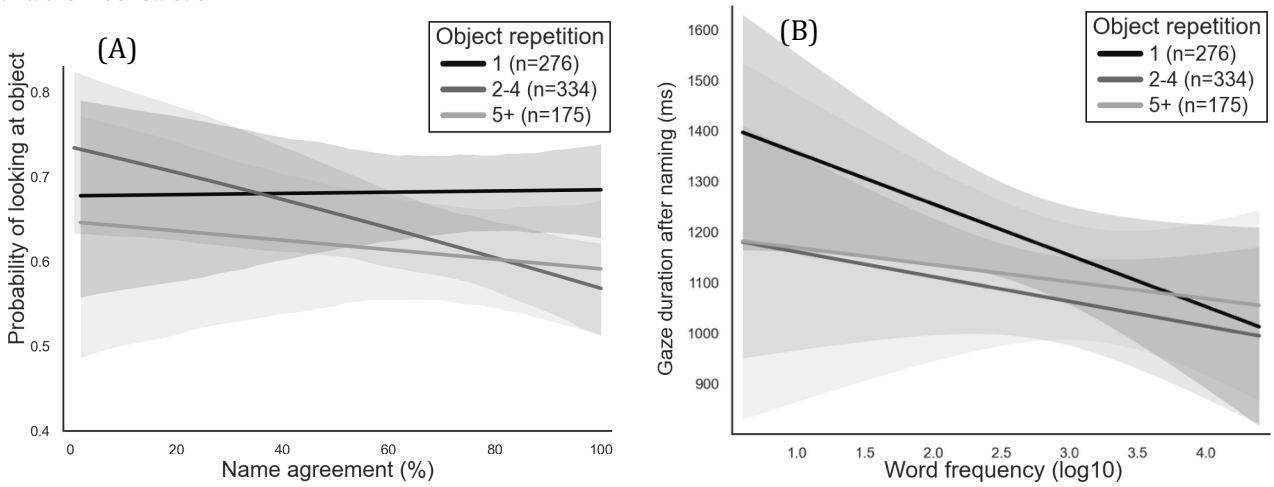

Figure 3. Study 2. (A) The mean probability (95% CI) of looking at objects after naming them as a function of name agreement and repetition, collapsing over other variables for simplicity. (B) Mean gaze duration (95% CI) for looking at objects as a function of word frequency and repetition, collapsing over other variables for simplicity.

References: Dahan et al. (2001) *Cogn. Psychol.*; Griffin (2001) *Cognition*; Levelt (1983) *Cognition*; Meyer et al. (1998) *Cognition*; Meyer & Van der Meulen (2000) *Psychon. Bull. Rev.*